

Pre-work · Reproductive Systems and Pregnancy

The Male Reproductive System, The Female Reproductive System, and Pregnancy and Birth. Read all three Part A chapters first. The two tracts are built on the same plan, a gonad, a duct system, accessory glands, and an outlet, so build them side by side and the third chapter reads as the sequel to both.

Module 4 · 4 of 5

Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1 Start with the reading.

Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.

2 Organize what you read.

In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.

3 Draw it.

In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.

4 Apply it.

Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.

5 Come back to it later in the week

, not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

3 min

PANEL 1 OF 3

Turn the three reproductive chapters into mini visuals. Eight chunks. Two chains carry the two routes, and the second of them runs on past fertilization into the first week of a pregnancy.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 HUB

The testis, from tunic to tubule

One testis at the hub, cut open. Spoke to the tunica vaginalis on the outside, the tunica albuginea and the lobules it divides, the seminiferous tubules coiled inside one lobule, and the epididymis on the posterior surface. Then spoke to the two cell types, sustentacular cells lining the tubule and interstitial cells lying between the tubules, and write the job of each. Add the dartos and the cremaster around the outside, and mark which is smooth muscle and which is skeletal.

2 CHAIN

The path of sperm, tubule to the outside

Five boxes with arrows, seminiferous tubule through to the urethra. Under each box write whether that structure makes, stores, matures, or carries. Only the first box can say makes, and the chain gets easier to hold once you see that.

3 HUB

A sperm cell

One sperm at large scale. Spoke to the head with its nucleus and the acrosome capping it, then to the neck, middle piece, principal piece, and end piece along the tail. Write the one job of each part beside it, and mark the part packed with mitochondria.

4 GRID

The three accessory glands

Gland down the side, three rows. What it adds to the semen, where it empties, and one structural note across the top. Star the one gland the urethra runs straight through.

5 HUB

The ovary in section

One almond-shaped ovary at the hub, cut open. Spoke outward through the germinal epithelium and tunica albuginea to the cortex with its follicles and the medulla with its vessels. Draw a follicle at three sizes around the rim, then the mature follicle, the corpus luteum, and the corpus albicans. Hang the ovarian, suspensory, and broad ligaments off the outside.

6 CHAIN

From ovulation to implantation

Seven boxes with arrows, ovary through to the endometrium. Mark the box where fertilization normally happens and the box where implantation happens, and write above the chain that about a week of travelling and dividing separates the two.

7 LADDER

The three layers of the uterine wall

Three rungs, perimetrium at the top down to endometrium. Beside each write what it is made of and what it does. Split the bottom rung in two, stratum functionalis above stratum basalis, and mark which of the two is shed.

8 GRID

The temporary structures of pregnancy

Structure down the side, six rows. What it is and what it does across the top. Add a third column asking whether it is kept or discarded at birth. Every row answers the same way, and that is the point of the grid.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

3 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

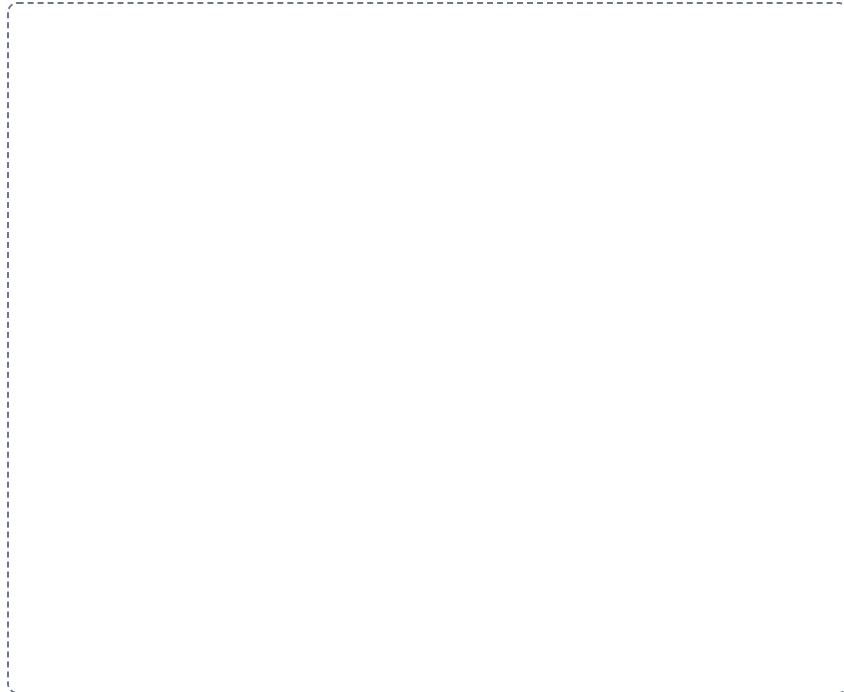
DRAWING 1

The male tract in sagittal section

BRAIN DUMP FIRST

Two minutes. Every duct and every gland in order, from the testis to the outside.

Draw the male pelvis cut down the midline, with the bladder, the pubic symphysis, and the rectum in place. Add the testis in the scrotum, the epididymis on its posterior surface, and the ductus deferens running up in the spermatic cord and down behind the bladder. Draw the seminal vesicle joining it to form the ejaculatory duct, the prostate wrapped around the urethra below the bladder, and the bulbourethral glands beneath that. Follow the urethra through its prostatic, membranous, and spongy parts, then draw the penis in cross section with the two corpora cavernosa and the single corpus spongiosum around the urethra.



In your notebook, head the page **M4-10 The male tract in sagittal section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can trace the whole route with a finger and say at every junction what has just been added to the sperm.

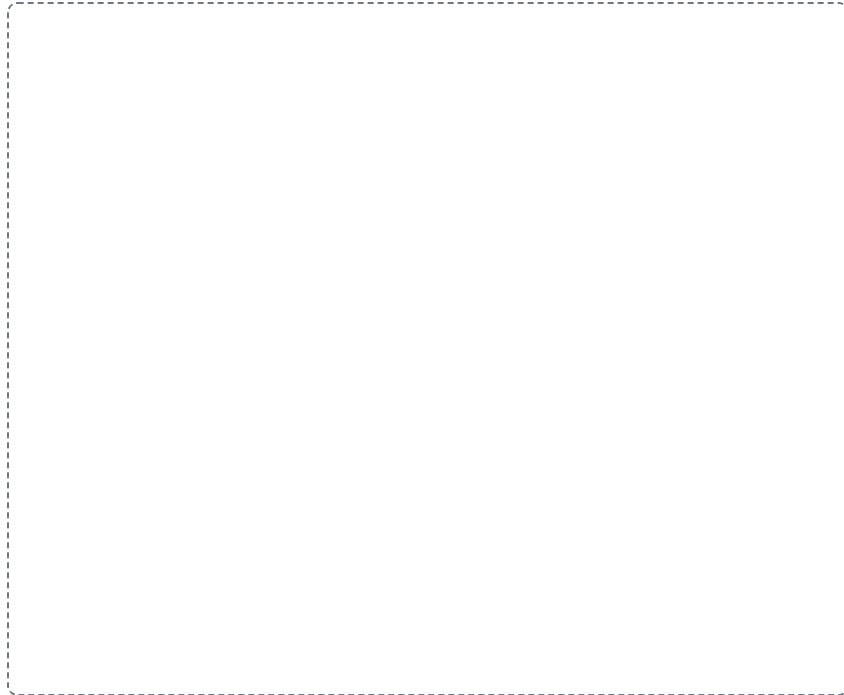
DRAWING 2

The uterus, tubes, and ovaries, in frontal view

BRAIN DUMP FIRST

Two minutes. Both tubes with every region, the uterus with every region, and the wall in layers.

Draw the uterus pear-shaped and cut open on one side. Mark the fundus, body, cervix, uterine cavity, and cervical canal with its internal and external os, and label the three wall layers along the cut edge. Draw both uterine tubes running out from the top and mark the isthmus, ampulla, infundibulum, and fimbriae on each, with the fimbriae reaching over the ovary rather than sealing onto it. Add an ovary at each end and draw the ovarian, suspensory, broad, round, cardinal, and uterosacral ligaments where each one attaches.



In your notebook, head the page **M4-11 The uterus, tubes, and ovaries, in frontal view** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain from your own drawing how an infection reaches the pelvic cavity from the vagina, and why that open route exists at all.

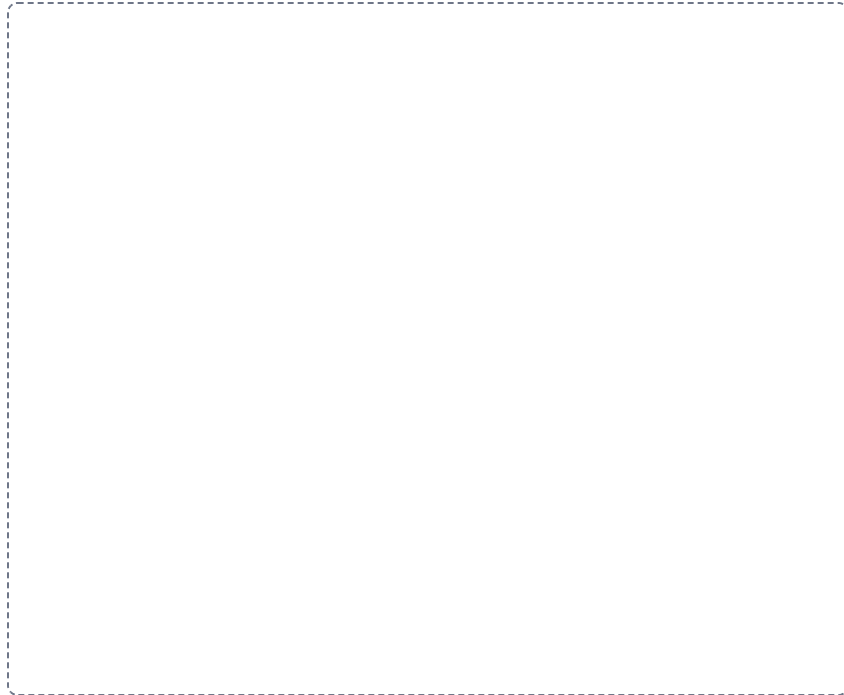
DRAWING 3

The uterus at full term, and the three stages of labor

BRAIN DUMP FIRST

Two minutes. The fetus in position, every temporary structure, and the order of birth.

Draw the gravid uterus in sagittal section, reaching up toward the rib cage, with the fetus in the vertex position, head down at the cervix. Draw the placenta on the uterine wall, the umbilical cord with its two arteries and one vein, the amnion holding the amniotic fluid around the fetus, and the chorion outside that. Then draw three small panels beside it, one for each stage of labor: the cervix thinning and dilating, the fetus being pushed through the cervix and vagina, and the placenta delivered as the afterbirth.



In your notebook, head the page **M4-12 The uterus at full term, and the three stages of labor** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can name every structure that leaves the body at birth and say which of them was built for this pregnancy alone.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A patient with a history of pelvic inflammatory disease has severe pain on one side of the pelvis and a positive pregnancy test, and no embryo is seen in the uterus. Name the most likely site and explain the prediction from structure.

Answer. The ampulla of the uterine tube.

Reasoning. Start with where fertilization normally happens rather than with the pain. A sperm meets the oocyte in the ampulla, the widest and longest part of the tube, and the fertilized ovum then spends about a week travelling toward the uterus while it divides into a morula and then a blastocyst. Anything that slows that journey leaves the blastocyst still inside the tube at the moment it is ready to implant, and scarring left by an earlier infection is exactly that kind of delay. Implantation is not choosy about the site, only about the timing, so the embryo embeds wherever it happens to be, which rules out the idea that it must reach the endometrium first. The ampulla is where it usually is at that moment, and that is why it is the commonest site. A tube a few millimeters wide has no lining built to stretch, which is what makes this a structural emergency rather than a merely misplaced pregnancy.

Set A • Name it

1. The cells lying between the seminiferous tubules that produce testosterone are the _____.
2. The enzyme-filled cap over the head of a sperm cell is the _____.
3. The widest and longest region of the uterine tube, where fertilization usually occurs, is the _____.
4. The outer layer of the blastocyst, which becomes the fetal part of the placenta, is the _____.
5. Name the three stages of labor in order and say what ends each one.

Set B • Predict it

1. The prostate enlarges. Predict the symptom and explain it from the path of the urethra.

2. A testis fails to descend into the scrotum. Predict the effect on sperm production and say why.

3. The ductus deferens is cut and tied on both sides. Predict what is still present in the semen and what is missing.

4. The placenta forms over the opening of the cervix. Predict the problem at delivery and name the condition.

Set C • Tell it apart

1. Distinguish the dartos from the cremaster muscle by tissue type and by what each one moves.

2. Distinguish the stratum functionalis from the stratum basalis of the endometrium by what happens to each one every cycle.

3. Compare an embryo with a fetus by the weeks each name covers and by what is happening to the body in each period.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

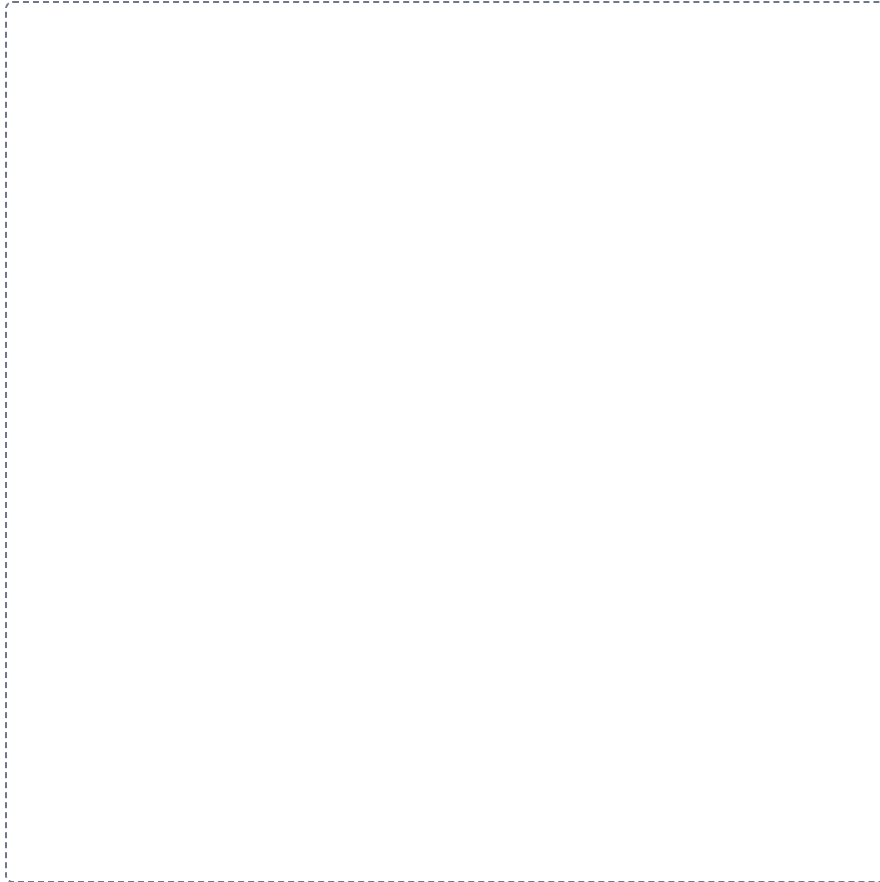
Mini visual sheet**2 min**

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

The testis, from tunic to tubule

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

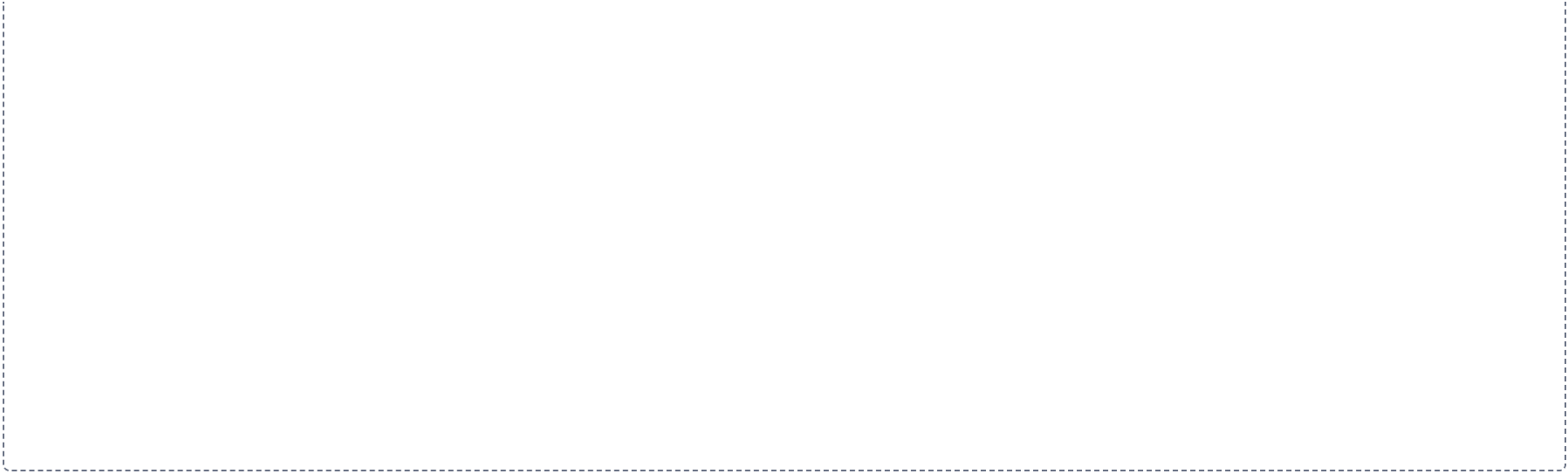


The path of sperm, tubule to the outside

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.

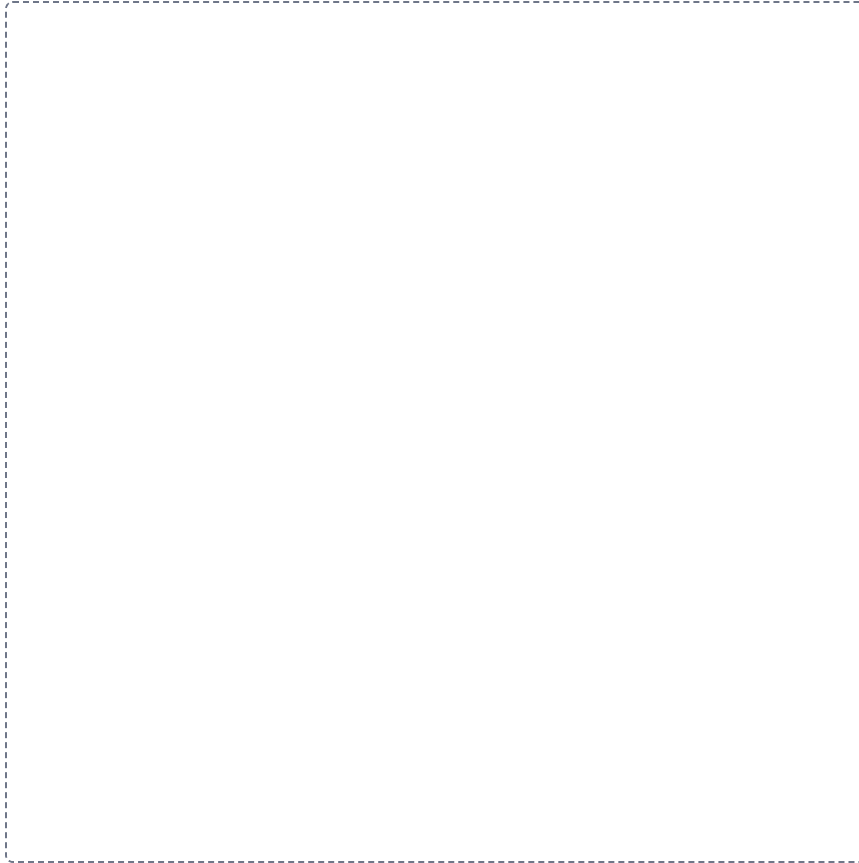
A large dashed rectangular box intended for a flowchart. The student is to draw the steps of the path of sperm from left to right, connecting them with arrows. Each step must be labeled, and each arrow must contain text describing the process that leads to the next step.



A sperm cell

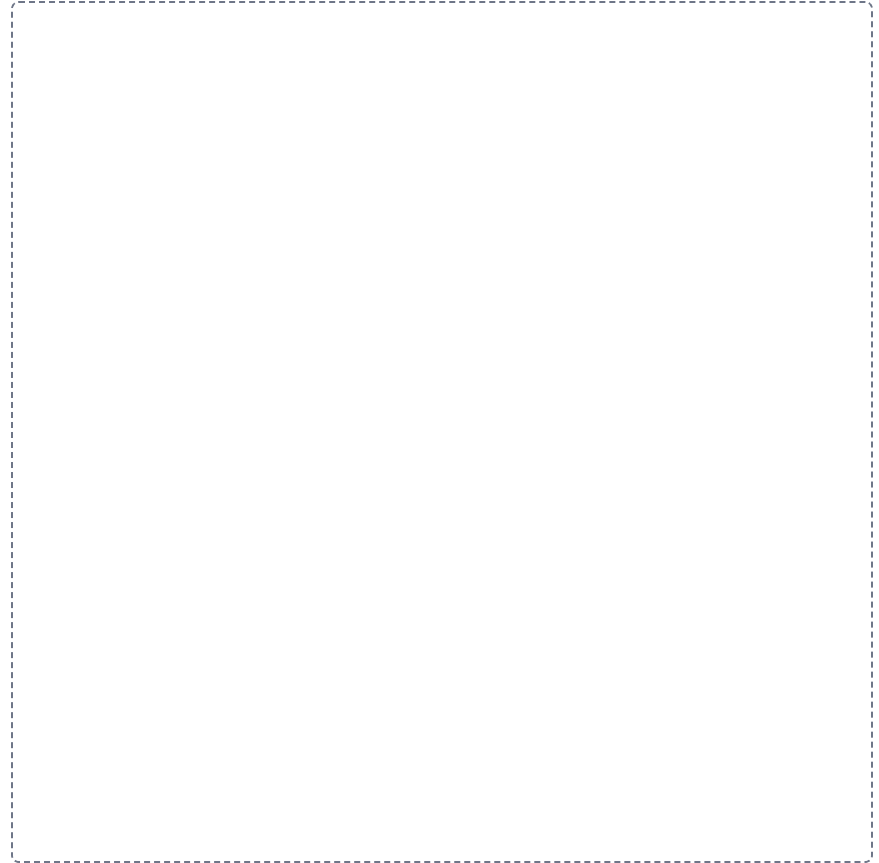
HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

**The three accessory glands**

GRID

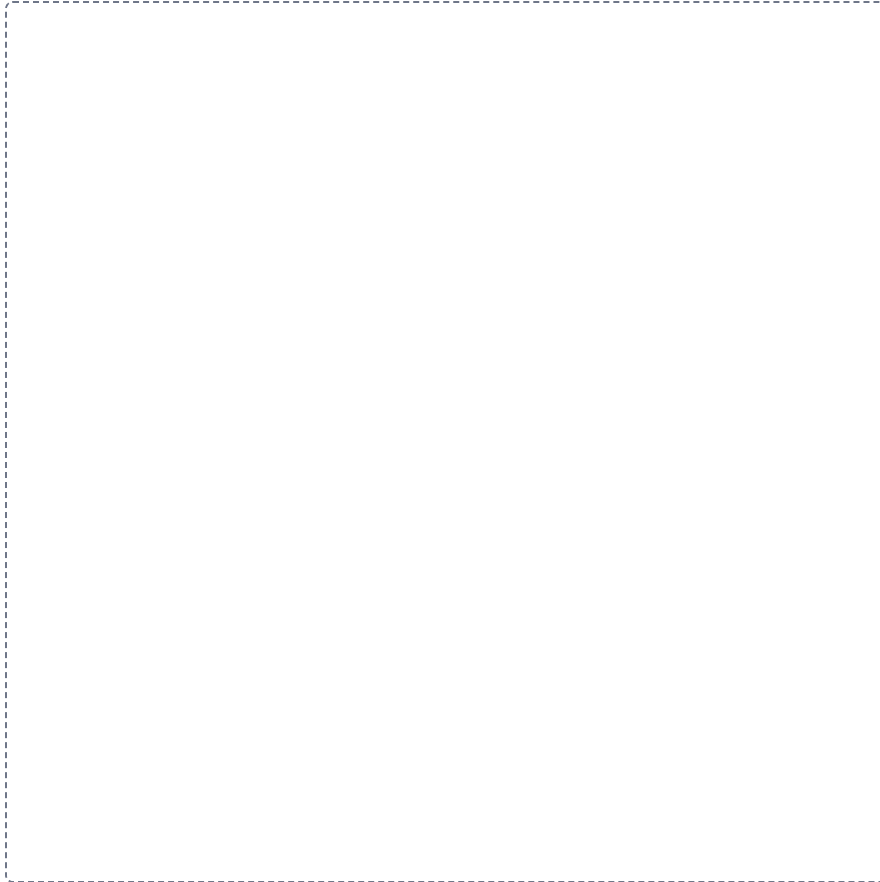
Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.



The ovary in section

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

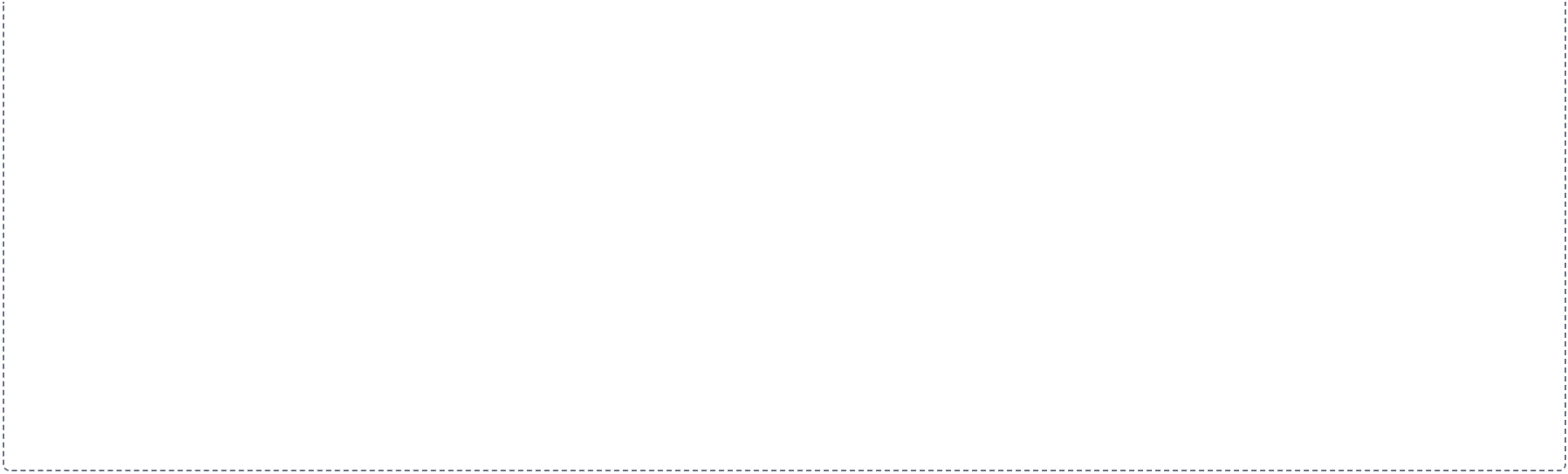


From ovulation to implantation

CHAIN

Draw the steps left to right with an arrow between each one. Label every step, and on each arrow write what happens to get to the next.



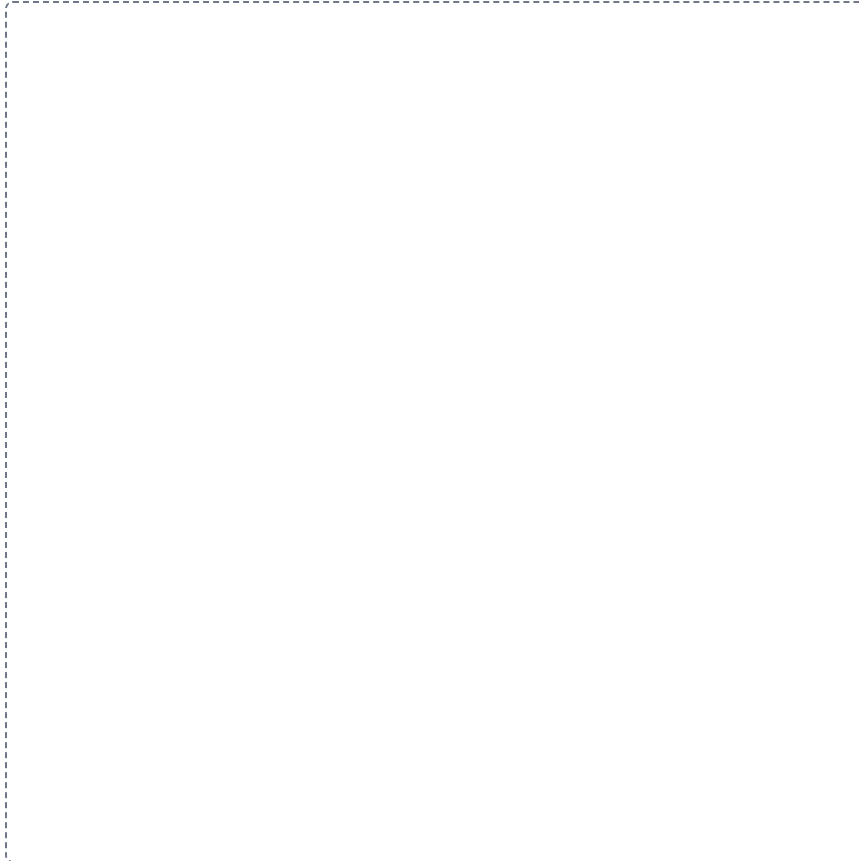


14-2

The three layers of the uterine wall

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.



14-3

The temporary structures of pregnancy

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

